

[+] Communicating with PM3 over USB-CDC

```
88888888b. 888b      d888      .d8888b.
888      Y88b 8888b      d8888      d88P      Y88b
888      888 888888b.d88888      .d88P
888      d88P 888Y888888P888      8888"
888888888P" 888 Y888P 888      "Y8b.
888      888 Y8P 888 888      888
888      888 "      888 Y88b d88P
888      888      888 "Y8888P"
```

[This is rather distressing! ☹]

[Proxmark3]

```
MCU..... AT91SAM7S512 Rev A
Memory.... 512 KB ( 77% used )
Target.... RDV4
```

```
Client.... Iceman/master/v4.20728-118-g3bdb7657f-dirty 2025-11-11 09:24:38
Bootrom... Iceman/master/v4.20728-118-g3bdb7657f-dirty-suspect 2025-11-26 17:08:34 15f0e45a6
OS..... Iceman/master/v4.20728-118-g3bdb7657f-dirty-suspect 2025-11-26 17:08:44 15f0e45a6
```

[usb] pm3 --> hf mfu info

```
[=] --- Tag Information -----
[+] TYPE: MIFARE Ultralight EV1 48bytes (MF0UL1101)
[+] UID: 04 86 67 7A AB 61 80
[+] UID[0]: 04, NXP Semiconductors Germany
[+] BCC0: 6D ( ok )
[+] BCC1: 30 ( ok )
[+] Internal: 48 ( default )
[+] Lock: 00 00 - 0000000000000000
[+] OTP: 00 00 00 00 - 00000000000000000000000000000000
```

```
[=] --- Tag Silicon Information
[+] Wafer Counter: 17577327 ( 0x10C356F )
[+] Wafer Coordinates: x 134, y 103 (0x86, 0x67)
[+] Test Site: 2
```

```
[=] --- Tag Counters
[+] [0]: 00 00 00
[+] - BD tearing ( ok )
[+] [1]: 00 00 00
[+] - BD tearing ( ok )
[+] [2]: 00 00 00
[+] - BD tearing ( ok )
```

```
[=] IC signature public key name: NXP Ultralight EV1
[=] IC signature public key value: 0490933BDCD6E99B4E255E3DA55389A8
[=] : 27564E11718E017292FAF23226A96614
[=] : B8
[=] Elliptic curve parameters: secp128r1
[=] TAG IC Signature: 9AA016CD5A42ECF73258911D1852BF57
[=] : C7FC74988DD312ACF34FB0BAEF609717
[+] Signature verification: successful
```

```
[=] --- Tag Version
[+] Raw bytes: 0004030101000B03
[+] Vendor ID: 04, NXP Semiconductors Germany
[+] Product type: Ultralight
[+] Product subtype: 01, 17 pF
[+] Major version: 01
[+] Minor version: 00
[+] Size: 0B, (64 - 32 bytes)
[+] Protocol type: 03, ISO14443-3 Compliant
```

```
[=] --- Tag Configuration
[+] cfg0 [16/0x10]: 000000FF
[+] - strong modulation mode disabled
[+] - pages don't need authentication
[+] cfg1 [17/0x11]: 00050000
[+] - Unlimited password attempts
[+] - NFC counter disabled
[+] - NFC counter not protected
[+] - user configuration writeable
[+] - write access is protected with password
[+] - 05, Virtual Card Type Identifier is default
[+] PWD [18/0x12]: 00000000 ( cannot be read )
[+] PACK [19/0x13]: 0000 ( cannot be read )
[+] RFU [19/0x13]: 0000 ( cannot be read )
```

```
[+] --- Known EV1/NTAG passwords
[+] Password... FFFFFFFF pack... 0000
```

```
[=] --- Fingerprint
[+] n/a
```

```
[usb] pm3 --> hf mfu dump
[+] TYPE: MIFARE Ultralight EV1 48bytes (MF0UL1101)
[+] Reading tag memory...
```

[=] MFU dump file information

```
[=] -----
[=] Version..... 00 04 03 01 01 00 0B 03
[=] TBD 0..... 00 00
[=] TBD 1..... 00
[=] Signature... 9A A0 16 CD 5A 42 EC F7 32 58 91 1D 18 52 BF 57
[=]              C7 FC 74 98 8D D3 12 AC F3 4F B0 BA EF 60 97 17
[=] Counter 0... 00 00 00
[=] Tearing 0... BD
[=] Counter 1... 00 00 00
[=] Tearing 1... BD
[=] Counter 2... 00 00 00
[=] Tearing 2... BD
[=] Max data page... 19 ( 80 bytes )
[=] Header size..... 56 bytes
```

```
[=] -----
[=] block# | data | lckl | ascii
[=] -----+-----+-----+-----
[=] 0/0x00 | 04 86 67 6D | | | ..gm
[=] 1/0x01 | 7A AB 61 80 | | | z?a.
[=] 2/0x02 | 30 48 00 00 | | | 0H..
[=] 3/0x03 | 00 00 00 00 | 0 | | ....
[=] 4/0x04 | 90 40 9D 5B | 0 | | .@.[
[=] 5/0x05 | 00 00 00 00 | 0 | | ....
[=] 6/0x06 | FF FF 00 00 | 0 | | ....
[=] 7/0x07 | 01 00 09 90 | 0 | | ....
[=] 8/0x08 | 9A F0 7D AE | 0 | | .?}?
[=] 9/0x09 | F2 54 41 15 | 0 | | ?TA.
[=] 10/0x0A | 00 00 00 00 | 0 | | ....
[=] 11/0x0B | 00 00 00 00 | 0 | | ....
[=] 12/0x0C | 00 00 00 00 | 0 | | ....
[=] 13/0x0D | 01 00 FF 10 | 0 | | ....
[=] 14/0x0E | 29 58 9D 5B | 0 | | )X.[
[=] 15/0x0F | 02 7F E5 12 | 0 | | .??.
[=] 16/0x10 | 00 00 00 FF | ? | | ....
[=] 17/0x11 | 00 05 00 00 | ? | | ....
[=] 18/0x12 | 00 00 00 00 | ? | | ....
[=] 19/0x13 | 00 00 00 00 | ? | | ....
[=] -----
```

```
[=] Using UID as filename
[+] Saved 136 bytes to binary file `./Users/wangzhenghong/hf-mfu-0486677AAB6180-dump.bin`
[+] Saved to json file ./Users/wangzhenghong/hf-mfu-0486677AAB6180-dump.json
[usb] pm3 --> hf 14a sniff
[=] Press pm3 button to abort sniffing
```

```
[#] trace len = 0
[usb] pm3 --> hf 14a sniff
[=] Press pm3 button to abort sniffing
```

```
[#] trace len = 3763
[usb] pm3 --> hf mfu list
[+] Recorded activity ( 3763 bytes )
[=] start = start of start frame. end = end of frame. src = source of transfer.
[=] ISO14443A - all times are in carrier periods (1/13.56MHz)
```

Start	End	Src	Data (! denotes parity error)	CRC	Annotation
0	1056	Rdr	26(7)		REQA
2260	2708	Tag	04(2)		
183392	184448	Rdr	26(7)		REQA
185652	186100	Tag	04(2)		
366336	367392	Rdr	26(7)		REQA
368580	370948	Tag	44 00		
513632	514688	Rdr	26(7)		REQA
696800	697856	Rdr	26(7)		REQA
699044	701412	Tag	44 00		
879632	880688	Rdr	26(7)		REQA
1043264	1044320	Rdr	26(7)		REQA
1045508	1047876	Tag	44 00		
1226976	1228032	Rdr	26(7)		REQA
1410112	1411168	Rdr	26(7)		REQA
1412372	1414740	Tag	44 00		
1877664	1878720	Rdr	26(7)		REQA
2061136	2062192	Rdr	26(7)		REQA
2244048	2245104	Rdr	26(7)		REQA
2408096	2409152	Rdr	26(7)		REQA
2591200	2592256	Rdr	26(7)		REQA
2774272	2775328	Rdr	26(7)		REQA
2937984	2939040	Rdr	26(7)		REQA
2974304	2976768	Rdr	93 20		ANTICOLL
3000640	3011168	Rdr	93 70 88 04 86 67 6D 8E 31	ok	SELECT_UID
3032512	3034976	Rdr	95 20		ANTICOLL-2
3058688	3069216	Rdr	95 70 7A AB 61 80 30 FC 58	ok	SELECT_UID-2
3568304	3571856	Rdr	C2 E0 B4	ok	
4041456	4042512	Rdr	26(7)		REQA
4043700	4046068	Tag	44 00		
4076752	4079216	Rdr	93 20		ANTICOLL
4080404	4086292	Tag	88 04 86 67 6D		
4114756	4118276	Tag	04 DA 17	ok	
4134880	4137344	Rdr	95 20		ANTICOLL-2
4138532	4144356	Tag	7A AB 61 80 30		
4161056	4171584	Rdr	95 70 7A AB 61 80 30 FC 58	ok	SELECT_UID-2
4172772	4176356	Tag	00 FE 51	ok	
4509920	4514624	Rdr	30 04 26 EE	ok	READBLOCK(4)
4515876	4536676	Tag	90 40 9D 5B 00 00 00 00 FF FF 00 00 01 00 09 90 99 5C	ok	
5006676	5027540	Tag	04 86 67 6D 7A AB 61 80 30 48 00 00 00 00 00 00 91 2A	ok	

17211524	17232388	Tag	04	86	67	6D	7A	AB	61	80	30	48	00	00	00	00	00	00	91	2A	ok		
17242340	17263140	Tag	7A	AB	61	80	30	48	00	00	00	00	00	00	90	40	9D	5B	7C	36	ok		
17272804	17293604	Tag	30	48	00	00	00	00	00	00	90	40	9D	5B	00	00	00	00	A1	7E	ok		
17297248	17301952	Rdr	30	03	99	9A															ok	READBLOCK(3)	
17303204	17324004	Tag	00	00	00	00	90	40	9D	5B	00	00	00	00	FF	FF	00	00	07	09	ok		
17889264	17893968	Rdr	30	04	26	EE															ok	READBLOCK(4)	
17895220	17916020	Tag	90	40	9D	5B	00	00	00	00	FF	FF	00	00	01	00	09	90	99	5C	ok		
17919920	17924624	Rdr	30	05	AF	FF															ok	READBLOCK(5)	
17925876	17946740	Tag	00	00	00	00	FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	52	FE	ok		
17950368	17955136	Rdr	30	06	34	CD															ok	READBLOCK(6)	
17956324	17977188	Tag	FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	F2	54	41	15	8E	0E	ok		
17980768	17985536	Rdr	30	07	BD	DC															ok	READBLOCK(7)	
17986724	18007588	Tag	01	00	09	90	9A	F0	7D	AE	F2	54	41	15	00	00	00	00	D4	46	ok		
18440416	18445120	Rdr	30	08	4A	24															ok	READBLOCK(8)	
18446372	18467172	Tag	9A	F0	7D	AE	F2	54	41	15	00	00	00	00	00	00	00	00	A0	05	ok		
18477092	18497892	Tag	1F2	54	41	15	00	00	00	00	00	00	00	00	00	00	00	00	3B	60	ok		
18501584	18506352	Rdr	30	0A	58	07															ok	READBLOCK(10)	
18507540	18528404	Tag	00	00	00	00	00	00	00	00	00	00	00	00	01	00	FF	10	CD	BA	ok		
18537956	18558820	Tag	00	00	00	00	00	00	00	00	01	00	FF	10	29	58	9D	5B	80	F7	ok		
19108196	19129060	Tag	00	00	00	00	00	00	00	00	00	00	00	00	00	01	00	FF	10	CD	BA	ok	
19134688	19136000	Rdr	ICF!																				
19138916	19159780	Tag	00	00	00	00	00	00	00	00	01	00	FF	10	29	58	9D	5B	80	F7	ok		
19163392	19168160	Rdr	30	0C	6E	62															ok	READBLOCK(12)	
19169348	19190148	Tag	00	00	00	00	01	00	FF	10	29	58	9D	5B	02	7F	E5	12	08	B2	ok		
19802432	19807200	Rdr	30	0D	E7	73															ok	READBLOCK(13)	
19808388	19829188	Tag	01	00	FF	10	29	58	9D	5B	02	7F	E5	12	00	00	00	FF	D6	1D	ok		
19839172	19859972	Tag	29	58	9D	5B	02	7F	E5	12	00	00	00	FF	00	05	00	00	D6	B7	ok		
19869620	19890420	Tag	02	7F	E5	12	00	00	00	FF	00	05	00	00	00	00	00	00	79	78	ok		
20729908	20732276	Tag	44	00																			
20762768	20765232	Rdr	93	20																		ANTICOLL	
20766420	20772308	Tag	88	04	86	67	6D																
20789040	20799568	Rdr	93	70	88	04	86	67	6D	8E	31										ok	SELECT_UID	
20800756	20804276	Tag	04	DA	17																ok		
20820912	20823376	Rdr	95	20																		ANTICOLL-2	
20824564	20830388	Tag	7A	AB	61	80	30																
20847088	20857616	Rdr	95	70	7A	AB	61	80	30	FC	58										ok	SELECT_UID-2	
20858804	20862388	Tag	00	FE	51																ok		
21468368	21471920	Rdr	IC2	E0	B4																ok		
21631328	21634880	Rdr	IC2	E0	B4																ok		
21940768	21941824	Rdr	I26(7)																			REQA	
21943012	21945380	Tag	44	00																			
21979684	21985572	Tag	88	04	86	67	6D																
22002320	22012848	Rdr	93	70	88	04	86	67	6D	8E	31										ok	SELECT_UID	
22014036	22017556	Tag	04	DA	17																ok		
22034160	22036624	Rdr	95	20																		ANTICOLL-2	
22037812	22043636	Tag	7A	AB	61	80	30																
22060336	22070864	Rdr	95	70	7A	AB	61	80	30	FC	58										ok	SELECT_UID-2	
22072052	22075636	Tag	00	FE	51																ok		
22417188	22437988	Tag	90	40	9D	5B	00	00	00	00	FF	FF	00	00	01	00	09	90	99	5C	ok		
22907204	22928068	Tag	04	86	67	6D	7A	AB	61	80	30	48	00	00	00	00	00	00	91	2A	ok		
22932080	22936848	Rdr	30	01	8B	89															ok	READBLOCK(1)	
22938036	22958836	Tag	7A	AB	61	80	30	48	00	00	00	00	00	00	90	40	9D	5B	7C	36	ok		
22962496	22967200	Rdr	30	02	10	8B															ok	READBLOCK(2)	
22968452	22989252	Tag	30	48	00	00	00	00	00	00	90	40	9D	5B	00	00	00	00	A1	7E	ok		
22998804	23019604	Tag	00	00	00	00	90	40	9D	5B	00	00	00	00	FF	FF	00	00	07	09	ok		
23581892	23602692	Tag	90	40	9D	5B	00	00	00	00	FF	FF	00	00	01	00	09	90	99	5C	ok		
23606736	23611440	Rdr	30	05	AF	FF															ok	READBLOCK(5)	
23612692	23633556	Tag	00	00	00	00	FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	52	FE	ok		
23637184	23641952	Rdr	30	06	34	CD															ok	READBLOCK(6)	
23643140	23664004	Tag	FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	F2	54	41	15	8E	0E	ok		
23667584	23672352	Rdr	30	07	BD	DC															ok	READBLOCK(7)	
23673540	23694404	Tag	01	00	09	90	9A	F0	7D	AE	F2	54	41	15	00	00	00	00	D4	46	ok		
24277120	24281824	Rdr	30	08	4A	24															ok	READBLOCK(8)	
24283076	24303876	Tag	9A	F0	7D	AE	F2	54	41	15	00	00	00	00	00	00	00	00	A0	05	ok		
24307872	24312576	Rdr	30	09	C3	35															ok	READBLOCK(9)	
24313828	24334628	Tag	1F2	54	41	15	00	00	00	00	00	00	00	00	00	00	00	00	3B	60	ok		
24338336	24343104	Rdr	30	0A	58	07															ok	READBLOCK(10)	
24344292	24365156	Tag	00	00	00	00	00	00	00	00	00	00	00	00	01	00	FF	10	CD	BA	ok		
24368704	24373472	Rdr	30	0B	D1	16															ok	READBLOCK(11)	
24374660	24395524	Tag	00	00	00	00	00	00	00	00	01	00	FF	10	29	58	9D	5B	80	F7	ok		
24816704	24821472	Rdr	30	0A	58	07															ok	READBLOCK(10)	
24822660	24823044	Tag	I00(2)																				
25477136	25481904	Rdr	30	0D	E7	73															ok	READBLOCK(13)	

[usb] pm3 --> hf mfu list
[+] Recorded activity (3763 bytes)
[=] start = start of frame. end = end of frame. src = source of transfer.
[=] ISO14443A - all times are in carrier periods (1/13.56MHz)

Start	End	Src	Data (! denotes parity error)	CRC	Annotation
0	1056	Rdr	I26(7)		REQA
2260	2708	Tag	I04(2)		
183392	184448	Rdr	I26(7)		REQA
185652	186100	Tag	I04(2)		
366336	367392	Rdr	I26(7)		REQA
368580	370948	Tag	I44 00		
513632	514688	Rdr	I26(7)		REQA
696800	697856	Rdr	I26(7)		REQA
699044	701412	Tag	I44 00		
879632	880688	Rdr	I26(7)		REQA
1043264	1044320	Rdr	I26(7)		REQA
1045508	1047876	Tag	I44 00		
1226976	1228032	Rdr	I26(7)		REQA
1410112	1411168	Rdr	I26(7)		REQA
1412372	1414740	Tag	I44 00		
1877664	1878720	Rdr	I26(7)		REQA

[illegible]

14308532	14329332	Tag	02	7F	E5	12	00	00	00	FF	00	05	00	00	00	00	00	79	78	ok	
15167520	15168576	Rdr	126(7)																		REQA
15169764	15172132	Tag	144	00																	
15202672	15205136	Rdr	193	20																	ANTICOLL
15206324	15212212	Tag	188	04	86	67	6D														
15228976	15239504	Rdr	193	70	88	04	86	67	6D	8E	31								ok		SELECT_UID
15240692	15244212	Tag	104	DA	17														ok		
15260832	15263296	Rdr	195	20																	ANTICOLL-2
15264484	15270308	Tag	17A	AB	61	80	30														
15287008	15297536	Rdr	195	70	7A	AB	61	80	30	FC	58								ok		SELECT_UID-2
15298724	15302308	Tag	100	FE	51														ok		
15907520	15911072	Rdr	1C2	E0	B4														ok		
15989344	15992896	Rdr	1C2	E0	B4														ok		
16070496	16074048	Rdr	1C2	E0	B4														ok		
16381844	16384212	Tag	144	00																	
16418500	16424388	Tag	188	04	86	67	6D														
16441152	16451680	Rdr	193	70	88	04	86	67	6D	8E	31								ok		SELECT_UID
16452868	16456388	Tag	104	DA	17														ok		
16476692	16482516	Tag	17A	AB	61	80	30														
16499216	16509744	Rdr	195	70	7A	AB	61	80	30	FC	58								ok		SELECT_UID-2
16510932	16514516	Tag	100	FE	51														ok		
16849120	16853824	Rdr	130	04	26	EE													ok		READBLOCK(4)
16855076	16875876	Tag	190	40	9D	5B	00	00	00	00	FF	FF	00	00	01	00	09	90	99	5C	ok
17211524	17232388	Tag	104	86	67	6D	7A	AB	61	80	30	48	00	00	00	00	00	00	91	2A	ok
17242340	17263140	Tag	17A	AB	61	80	30	48	00	00	00	00	00	00	90	40	9D	5B	7C	36	ok
17272804	17293604	Tag	130	48	00	00	00	00	00	00	90	40	9D	5B	00	00	00	00	A1	7E	ok
17297248	17301952	Rdr	130	03	99	9A													ok		READBLOCK(3)
17303204	17324004	Tag	100	00	00	00	90	40	9D	5B	00	00	00	00	FF	FF	00	00	07	09	ok
17889264	17893968	Rdr	130	04	26	EE													ok		READBLOCK(4)
17895220	17916020	Tag	190	40	9D	5B	00	00	00	00	FF	FF	00	00	01	00	09	90	99	5C	ok
17919920	17924624	Rdr	130	05	AF	FF													ok		READBLOCK(5)
17925876	17946740	Tag	100	00	00	00	FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	52	FE	ok
17950368	17955136	Rdr	130	06	34	CD													ok		READBLOCK(6)
17956324	17977188	Tag	1FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	F2	54	41	15	8E	0E	ok
17980768	17985536	Rdr	130	07	BD	DC													ok		READBLOCK(7)
17986724	18007588	Tag	101	00	09	90	9A	F0	7D	AE	F2	54	41	15	00	00	00	00	D4	46	ok
18440416	18445120	Rdr	130	08	4A	24													ok		READBLOCK(8)
18446372	18467172	Tag	19A	F0	7D	AE	F2	54	41	15	00	00	00	00	00	00	00	00	A0	05	ok
18477092	18497892	Tag	1F2	54	41	15	00	00	00	00	00	00	00	00	00	00	00	00	3B	60	ok
18501584	18506352	Rdr	130	0A	58	07													ok		READBLOCK(10)
18507540	18528404	Tag	100	00	00	00	00	00	00	00	00	00	00	00	01	00	FF	10	CD	BA	ok
18537956	18558820	Tag	100	00	00	00	00	00	00	00	00	01	00	FF	10	29	58	9D	5B	80	F7
19108196	19129060	Tag	100	00	00	00	00	00	00	00	00	00	00	00	01	00	FF	10	CD	BA	ok
19134688	19136000	Rdr	1CF																		
19138916	19159780	Tag	100	00	00	00	00	00	00	00	01	00	FF	10	29	58	9D	5B	80	F7	ok
19163392	19168160	Rdr	130	0C	6E	62													ok		READBLOCK(12)
19169348	19190148	Tag	100	00	00	00	01	00	FF	10	29	58	9D	5B	02	7F	E5	12	08	B2	ok
19802432	19807200	Rdr	130	0D	E7	73													ok		READBLOCK(13)
19808388	19829188	Tag	101	00	FF	10	29	58	9D	5B	02	7F	E5	12	00	00	00	FF	D6	1D	ok
19839172	19859972	Tag	129	58	0D	5B	02	7F	E5	12	00	00	00	FF	00	05	00	00	D6	B7	ok
19869620	19890420	Tag	102	7F	E5	12	00	00	00	FF	00	05	00	00	00	00	00	00	79	78	ok
20729908	20732276	Tag	144	00																	
20762768	20765232	Rdr	193	20																	ANTICOLL
20766420	20772308	Tag	188	04	86	67	6D														
20789040	20799568	Rdr	193	70	88	04	86	67	6D	8E	31								ok		SELECT_UID
20800756	20804276	Tag	104	DA	17														ok		
20820912	20823376	Rdr	195	20																	ANTICOLL-2
20824564	20830388	Tag	17A	AB	61	80	30														
20847088	20857616	Rdr	195	70	7A	AB	61	80	30	FC	58								ok		SELECT_UID-2
20858804	20862388	Tag	100	FE	51														ok		
21468368	21471920	Rdr	1C2	E0	B4														ok		
21631328	21634880	Rdr	1C2	E0	B4														ok		
21940768	21941824	Rdr	126(7)																		REQA
21943012	21945380	Tag	144	00																	
21979684	21985572	Tag	188	04	86	67	6D														
22002320	22012848	Rdr	193	70	88	04	86	67	6D	8E	31								ok		SELECT_UID
22014036	22017556	Tag	104	DA	17														ok		
22034160	22036624	Rdr	195	20																	ANTICOLL-2
22037812	22043636	Tag	17A	AB	61	80	30														
22060336	22070864	Rdr	195	70	7A	AB	61	80	30	FC	58								ok		SELECT_UID-2
22072052	22075636	Tag	100	FE	51														ok		
22417188	22437988	Tag	190	40	9D	5B	00	00	00	00	FF	FF	00	00	01	00	09	90	99	5C	ok
22907204	22928068	Tag	104	86	67	6D	7A	AB	61	80	30	48	00	00	00	00	00	00	91	2A	ok
22932080	22936848	Rdr	130	01	8B	B9													ok		READBLOCK(1)
22938036	22958836	Tag	17A	AB	61	80	30	48	00	00	00	00	00	00	90	40	9D	5B	7C	36	ok
22962496	22967200	Rdr	130	02	10	88													ok		READBLOCK(2)
22968452	22989252	Tag	130	48	00	00	00	00	00	00	90	40	9D	5B	00	00	00	00	A1	7E	ok
22998804	23019604	Tag	100	00	00	00	90	40	9D	5B	00	00	00	00	FF	FF	00	00	07	09	ok
23581892	23602692	Tag	190	40	9D	5B	00	00	00	00	FF	FF	00	00	01	00	09	90	99	5C	ok
23606736	23611440	Rdr	130	05	AF	FF													ok		READBLOCK(5)
23612692	23633556	Tag	100	00	00	00	FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	52	FE	ok
23637184	23641952	Rdr	130	06	34	CD													ok		READBLOCK(6)
23643140	23664004	Tag	1FF	FF	00	00	01	00	09	90	9A	F0	7D	AE	F2	54	41	15	8E	0E	ok
23667584	23672352	Rdr	130	07	BD	DC													ok		READBLOCK(7)
23673540	23694404	Tag	101	00	09	90	9A	F0	7D	AE	F2	54	41	15	00	00	00	00	D4	46	ok
24277120	24281824	Rdr	130	08	4A	24													ok		READBLOCK(8)
24283076	24303876	Tag	19A	F0	7D	AE	F2	54	41	15	00	00	00	00	00	00	00	00	A0	05	ok
24307872	24312576	Rdr	130																		

[usb] pm3 --> hf 14a sniff
[=] Press pm3 button to abort sniffing

[#] trace len = 4789
[usb] pm3 --> hf mfu list
[+] Recorded activity (4789 bytes)
[=] start = start of start frame. end = end of frame. src = source of transfer.
[=] ISO14443A - all times are in carrier periods (1/13.56MHz)

Start	End	Src	Data (! denotes parity error)	CRC	Annotation
0	384	Tag	00(2)		
183344	183728	Tag	00(2)		
531360	531552	Tag	01(0)		
714448	714704	Tag	00(1)		
1428112	1428496	Tag	00(2)		
1897600	1899968	Tag	44 00		
2263696	2263952	Tag	00(1)		
2609084	2610140	Rdr	26(7)		REQA
2792204	2793260	Rdr	26(7)		REQA
2953292	2954348	Rdr	26(7)		REQA
3137100	3138156	Rdr	26(7)		REQA
3320140	3321196	Rdr	26(7)		REQA
3794460	3795516	Rdr	26(7)		REQA
3977596	3978652	Rdr	26(7)		REQA
4160444	4161500	Rdr	26(7)		REQA
4323468	4324524	Rdr	26(7)		REQA
4506588	4507644	Rdr	26(7)		REQA
4689692	4690748	Rdr	26(7)		REQA
4853020	4854076	Rdr	26(7)		REQA
5036908	5037964	Rdr	26(7)		REQA
5220076	5221132	Rdr	26(7)		REQA
5691292	5692348	Rdr	26(7)		REQA
5726428	5728892	Rdr	93 20		ANTICOLL
5773996	5776460	Rdr	93 20		ANTICOLL
5822540	5825004	Rdr	93 20		ANTICOLL
5848860	5859324	Rdr	93 70 88 04 07 3D B6 67 21	ok	SELECT_UID
5904076	5914540	Rdr	93 70 88 04 07 3D B6 67 21	ok	SELECT_UID
5959692	5970156	Rdr	93 70 88 04 07 3D B6 67 21	ok	SELECT_UID
6133900	6134956	Rdr	26(7)		REQA
6168956	6171420	Rdr	93 20		ANTICOLL
6195244	6205708	Rdr	93 70 88 04 07 3D B6 67 21	ok	SELECT_UID
6227868	6230332	Rdr	95 20		ANTICOLL-2
6254140	6264604	Rdr	95 70 7A AB 61 81 31 AD 50	ok	SELECT_UID-2
6841356	6844908	Rdr	IC2 E0 B4	ok	
6923100	6926652	Rdr	IC2 E0 B4	ok	
7004236	7007788	Rdr	IC2 E0 B4	ok	
7317900	7318956	Rdr	26(7)		REQA
7353020	7355484	Rdr	93 20		ANTICOLL
7379324	7389788	Rdr	93 70 88 04 07 3D B6 67 21	ok	SELECT_UID
7411196	7413660	Rdr	95 20		ANTICOLL-2
7437372	7447836	Rdr	95 70 7A AB 61 81 31 AD 50	ok	SELECT_UID-2
7449296	7449488	Tag	01(0)		
7785276	7789980	Rdr	30 04 26 EE	ok	READBLOCK(4)
7791232	7812032	Tag	90 40 44 55 00 00 00 00 FF FF 00 00 01 00 09 90 F0 EE	ok	
8405500	8410268	Rdr	30 00 02 A8	ok	READBLOCK(0)
8411456	8432256	Tag	04 07 3D B6 7A AB 61 81 31 48 00 00 00 00 00 00 3F CA	ok	
8436348	8441116	Rdr	30 01 8B B9	ok	READBLOCK(1)
8442304	8463104	Tag	7A AB 61 81 31 48 00 00 00 00 00 00 90 40 44 55 D9 D7	ok	
8466828	8471532	Rdr	30 02 10 88	ok	READBLOCK(2)
8472784	8493648	Tag	31 48 00 00 00 00 00 00 90 40 44 55 00 00 00 00 B9 F1	ok	
8497212	8501916	Rdr	30 03 99 9A	ok	READBLOCK(3)
8503168	8524032	Tag	00 00 00 00 90 40 44 55 00 00 00 00 FF FF 00 00 03 B6	ok	
8948428	8953132	Rdr	30 04 26 EE	ok	READBLOCK(4)
8954384	8975184	Tag	90 40 44 55 00 00 00 00 FF FF 00 00 01 00 09 90 F0 EE	ok	
8985232	9006032	Tag	00 00 00 00 FF FF 00 00 01 00 09 90 CE 13 23 17 C0 DE	ok	
9009740	9014508	Rdr	30 06 34 CD	ok	READBLOCK(6)
9015696	9036496	Tag	FF FF 00 00 01 00 09 90 CE 13 23 17 99 AE 34 44 C3 B1	ok	
9040140	9044908	Rdr	30 07 BD DC	ok	READBLOCK(7)
9046096	9066896	Tag	01 00 00 00 CE 13 23 17 99 AE 34 44 00 00 00 00 69 1B	ok	
9636176	9656976	Tag	ICE 13 23 17 99 AE 34 44 00 00 00 00 00 00 00 00 2F 93	ok	
9666912	9687776	Tag	99 AE 34 44 00 00 00 00 00 00 00 00 00 00 00 00 79 7C	ok	
9691356	9696124	Rdr	30 0A 58 07	ok	READBLOCK(10)
9697312	9718176	Tag	00 00 00 00 00 00 00 00 00 00 00 00 01 00 FF 10 CD BA	ok	
9721740	9726508	Rdr	30 0B D1 16	ok	READBLOCK(11)
9727696	9748496	Tag	00 00 00 00 00 00 00 00 01 00 FF 10 02 49 44 55 D2 06	ok	
10320524	10325292	Rdr	30 0A 58 07	ok	READBLOCK(10)
10326480	10347344	Tag	00 00 00 00 00 00 00 00 00 00 00 00 01 00 FF 10 CD BA	ok	
10351228	10355996	Rdr	30 0B D1 16	ok	READBLOCK(11)
10357184	10377984	Tag	00 00 00 00 00 00 00 00 01 00 FF 10 02 49 44 55 D2 06	ok	
10381676	10386444	Rdr	30 0C 6E 62	ok	READBLOCK(12)
10387632	10408432	Tag	00 00 00 00 01 00 FF 10 02 49 44 55 02 7F E5 12 80 88	ok	
10871468	10876236	Rdr	30 0D E7 73	ok	READBLOCK(13)
10877424	10898288	Tag	01 00 FF 10 02 49 44 55 02 7F E5 12 00 00 00 FF B4 20	ok	
10902188	10906892	Rdr	30 0E 7C 41	ok	READBLOCK(14)
10908144	10929008	Tag	02 49 44 55 02 7F E5 12 00 00 00 FF 00 05 00 00 95 6E	ok	
10938576	10959376	Tag	02 7F E5 12 00 00 00 FF 00 05 00 00 00 00 00 00 79 78	ok	
12743500	12744556	Rdr	26(7)		REQA
12745744	12748112	Tag	44 00		
12782240	12788128	Tag	88 04 07 3D B6		
12804908	12815372	Rdr	93 70 88 04 07 3D B6 67 21	ok	SELECT_UID
12816624	12820144	Tag	04 DA 17	ok	
12836764	12839228	Rdr	95 20		ANTICOLL-2
12840416	12846304	Tag	7A AB 61 81 31		
12862940	12873404	Rdr	95 70 7A AB 61 81 31 AD 50	ok	SELECT_UID-2
12874656	12878240	Tag	00 FE 51	ok	

13349612	13353164	Rdr IC2 E0 B4																ok	
13512460	13516012	Rdr IC2 E0 B4																ok	
13825548	13826604	Rdr I26(7)																	REQA
13827792	13830160	Tag I44 00																	
13864336	13870224	Tag I88 04 07 3D B6																	
13898672	13902192	Tag I04 DA 17																ok	
13922464	13928352	Tag I7A AB 61 81 31																	
13944988	13955452	Rdr I95 70 7A AB 61 81 31 AD 50																ok	SELECT_UID-2
13956704	13960288	Tag I00 FE 51																ok	
14573184	14593984	Tag I90 40 44 55 00 00 00 FF FF 00 00 01 00 09 90 F0 EE																ok	
14922800	14943600	Tag I04 07 3D B6 7A AB 61 81 31 48 00 00 00 00 00 3F CA																ok	
14947548	14952316	Rdr I30 01 8B B9																ok	READBLOCK(1)
14953504	14974304	Tag I7A AB 61 81 31 48 00 00 00 00 00 00 40 44 55 D9 D7																ok	READBLOCK(2)
14977964	14982668	Rdr I30 02 10 8B																ok	
14983920	15004784	Tag I31 48 00 00 00 00 00 90 40 44 55 00 00 00 00 B9 F1																ok	
15008348	15013052	Rdr I30 03 99 9A																ok	READBLOCK(3)
15014304	15035168	Tag I00 00 00 00 90 40 44 55 00 00 00 FF FF 00 00 03 B6																ok	
15596160	15616960	Tag I90 40 44 55 00 00 00 FF FF 00 00 01 00 09 90 F0 EE																ok	
15626960	15647760	Tag I00 00 00 00 FF FF 00 00 01 00 09 90 CE 13 23 17 C0 DE																ok	
15657440	15678240	Tag IFF FF 00 00 01 00 09 90 CE 13 23 17 99 AE 34 44 C3 B1																ok	
15681868	15686636	Rdr I30 07 BD DC																ok	READBLOCK(7)
1568424	15708624	Tag I01 00 09 90 CE 13 23 17 99 AE 34 44 00 00 00 00 69 1B																ok	
161424824	16147132	Rdr I30 08 4A 24																ok	READBLOCK(8)
16148384	16169184	Tag ICE 13 23 17 99 AE 34 44 00 00 00 00 00 00 00 2F 93																ok	
16179184	16200048	Tag I99 AE 34 44 00 00 00 00 00 00 00 00 00 00 00 79 7C																ok	
16209648	16230512	Tag I00 00 00 00 00 00 00 00 00 00 00 01 00 FF 10 CD BA																ok	
16240064	16260864	Tag I00 00 00 00 00 00 00 00 01 00 FF 10 02 49 44 55 D2 06																ok	
16830460	16835228	Rdr I30 0A 58 07																ok	READBLOCK(10)
16836416	16857280	Tag I00 00 00 00 00 00 00 00 00 00 00 01 00 FF 10 CD BA																ok	
16861180	16865948	Rdr I30 0B D1 16																ok	READBLOCK(11)
16867136	16887936	Tag I00 00 00 00 00 00 00 00 01																	


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[+] BCC0: 41 ( OK )
[+] BCC1: 78 ( ok )
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